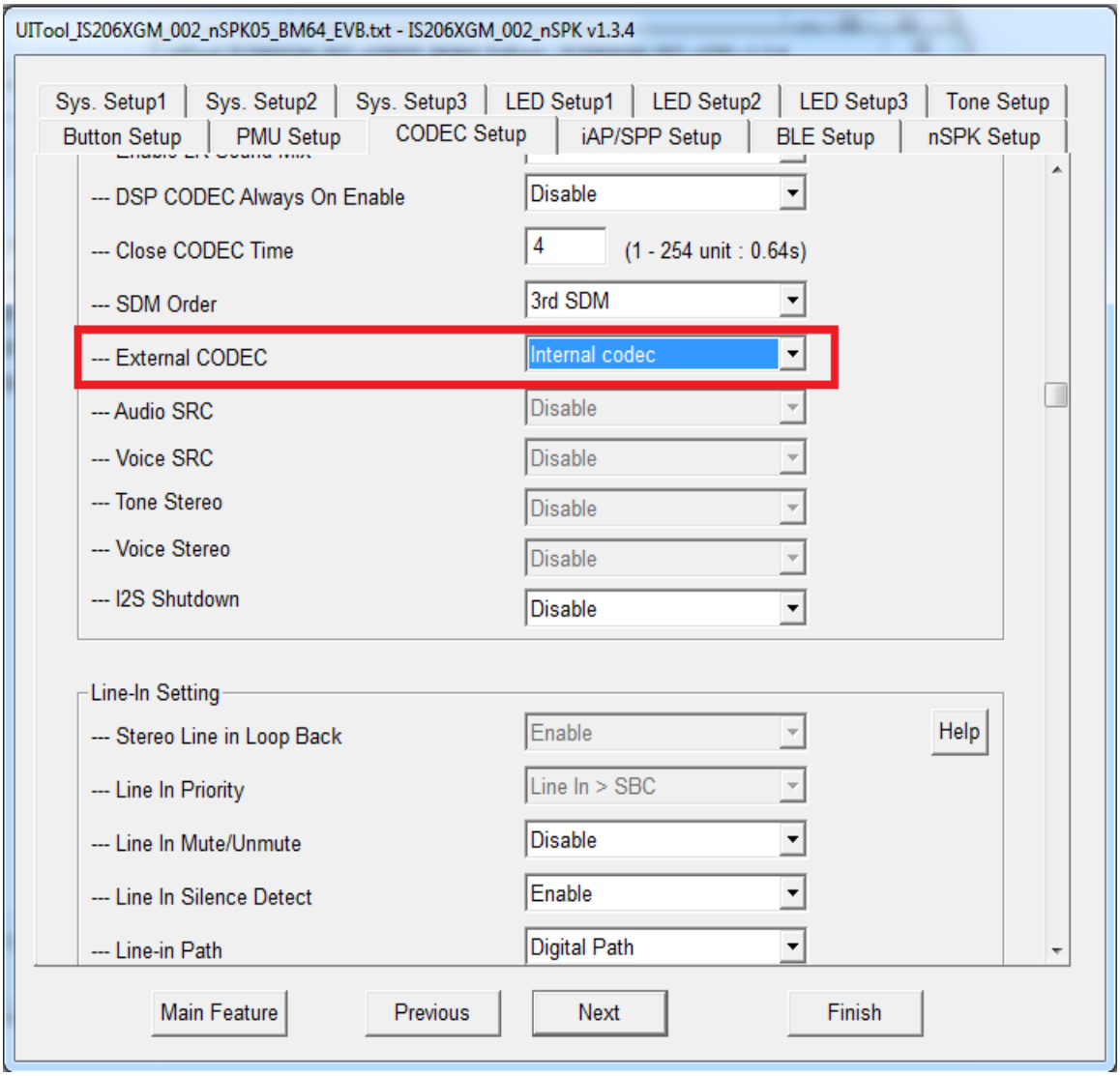
BM64 I2S/SPK out Configuration

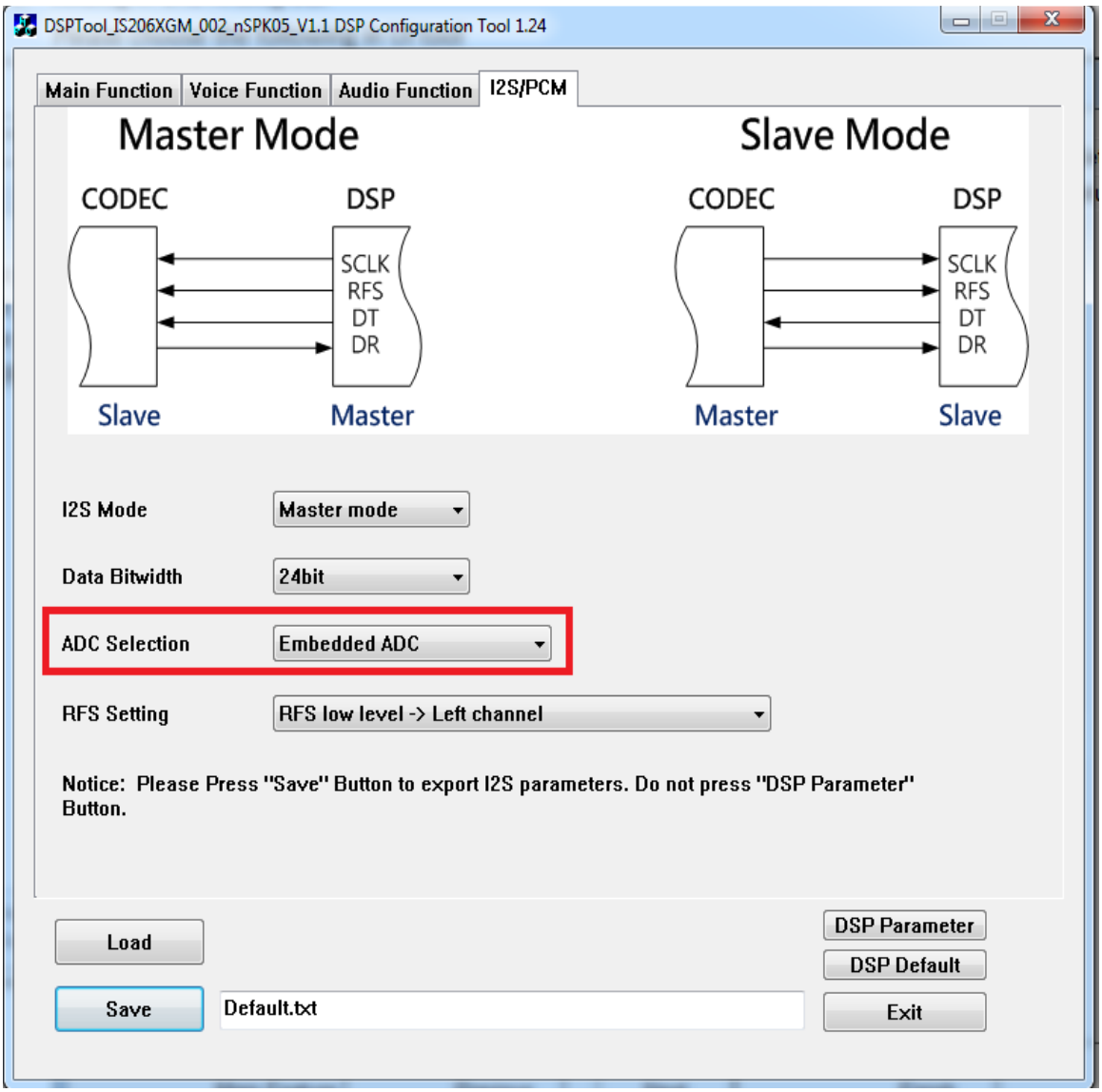
Aux-in/MIC in can be configured in BM64 in 4 possible combinations

1. **Analog in and Analog out**

   Please choose the following in UI tool

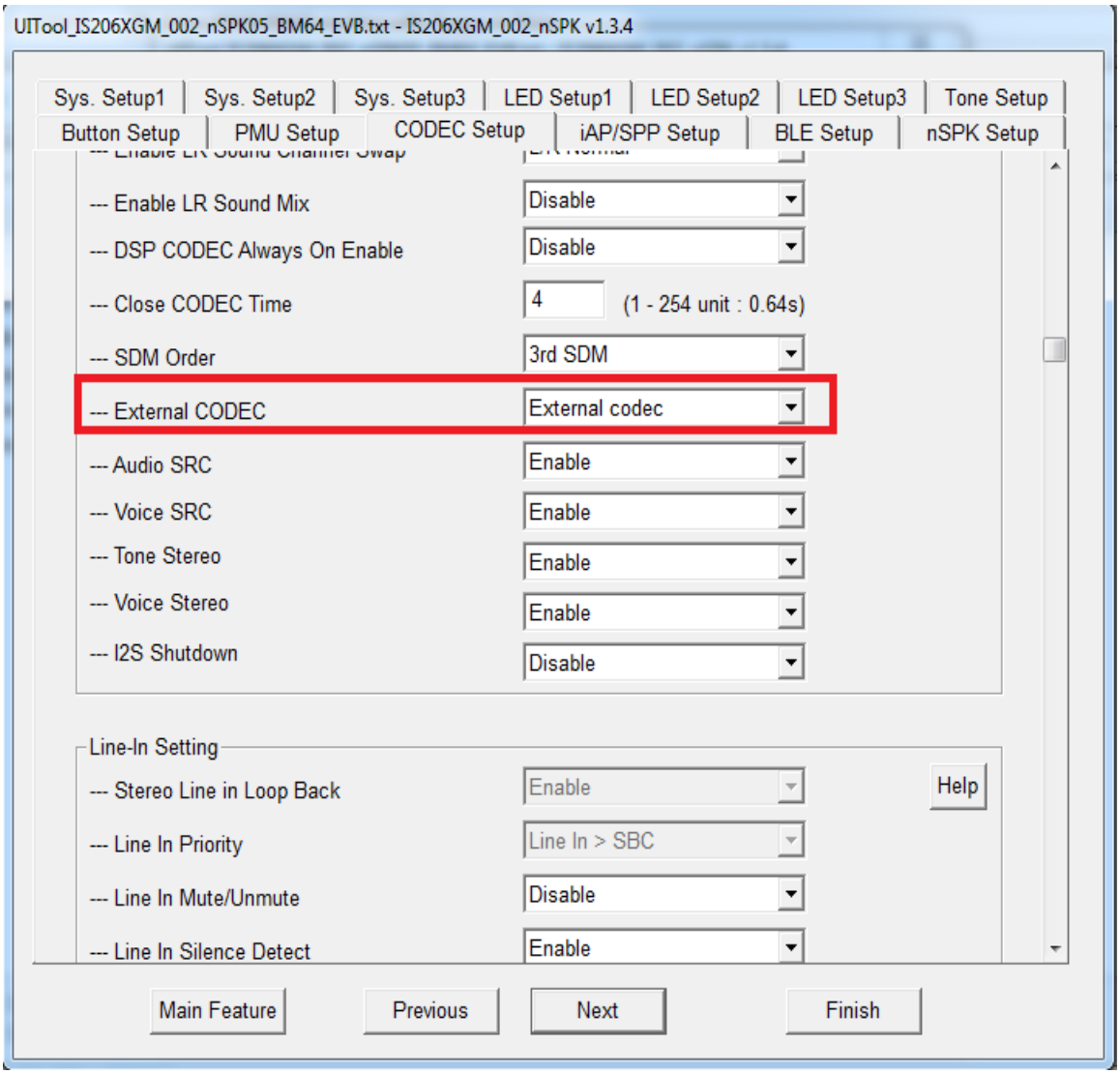


And following in DSP tool

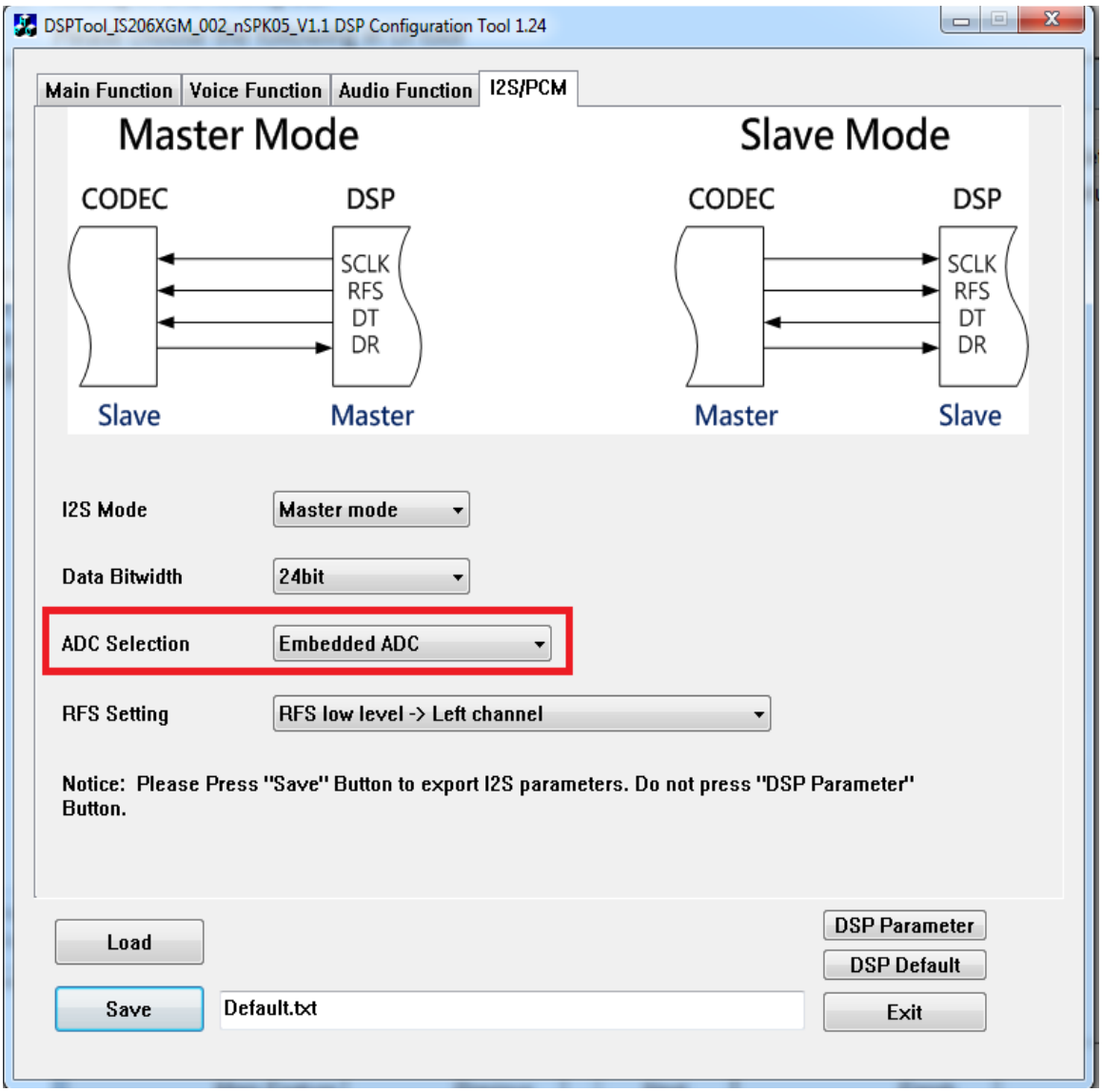


2. Analog in and I2S out

Please choose following in UI tool

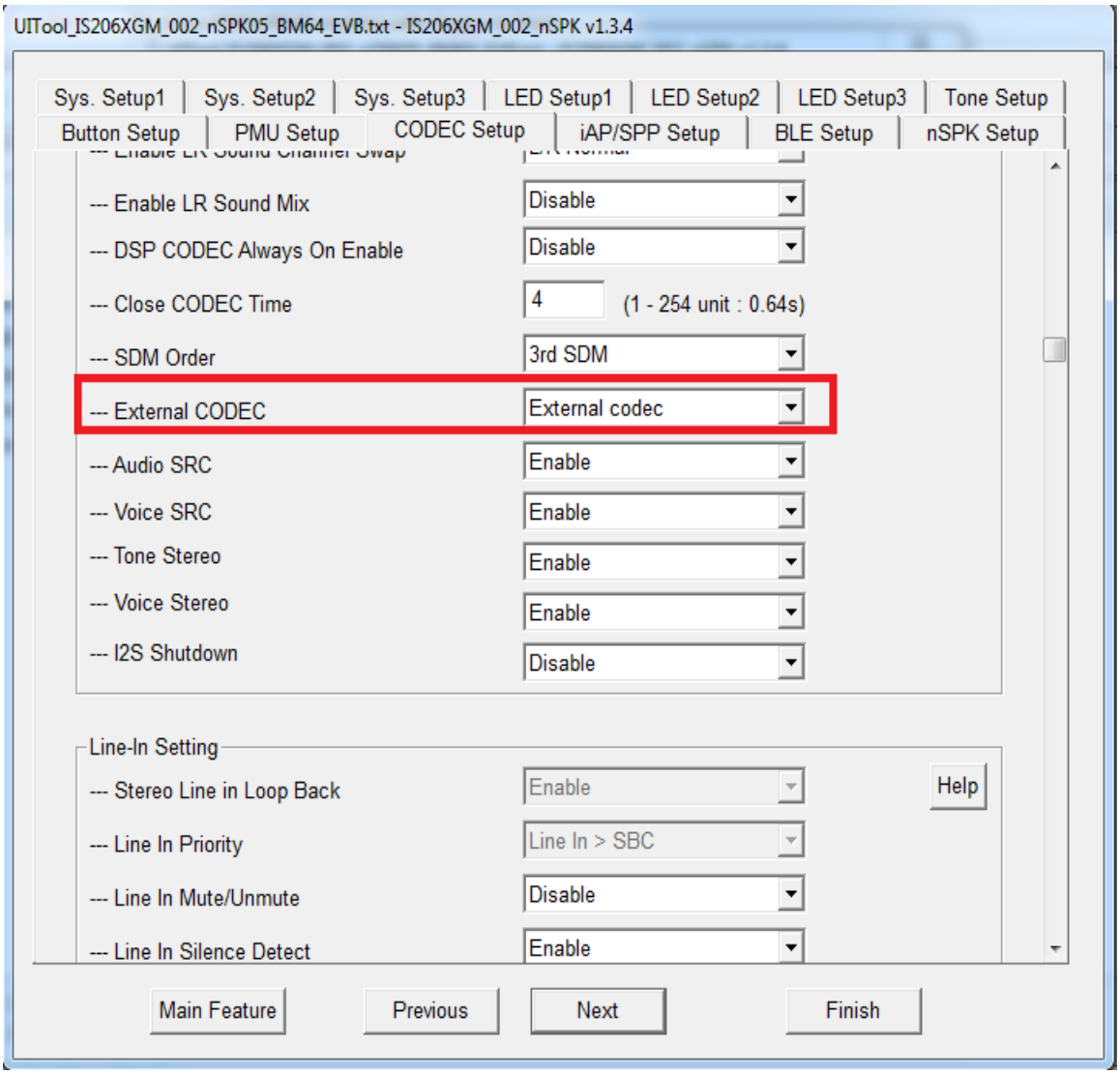


And following in DSP Tool

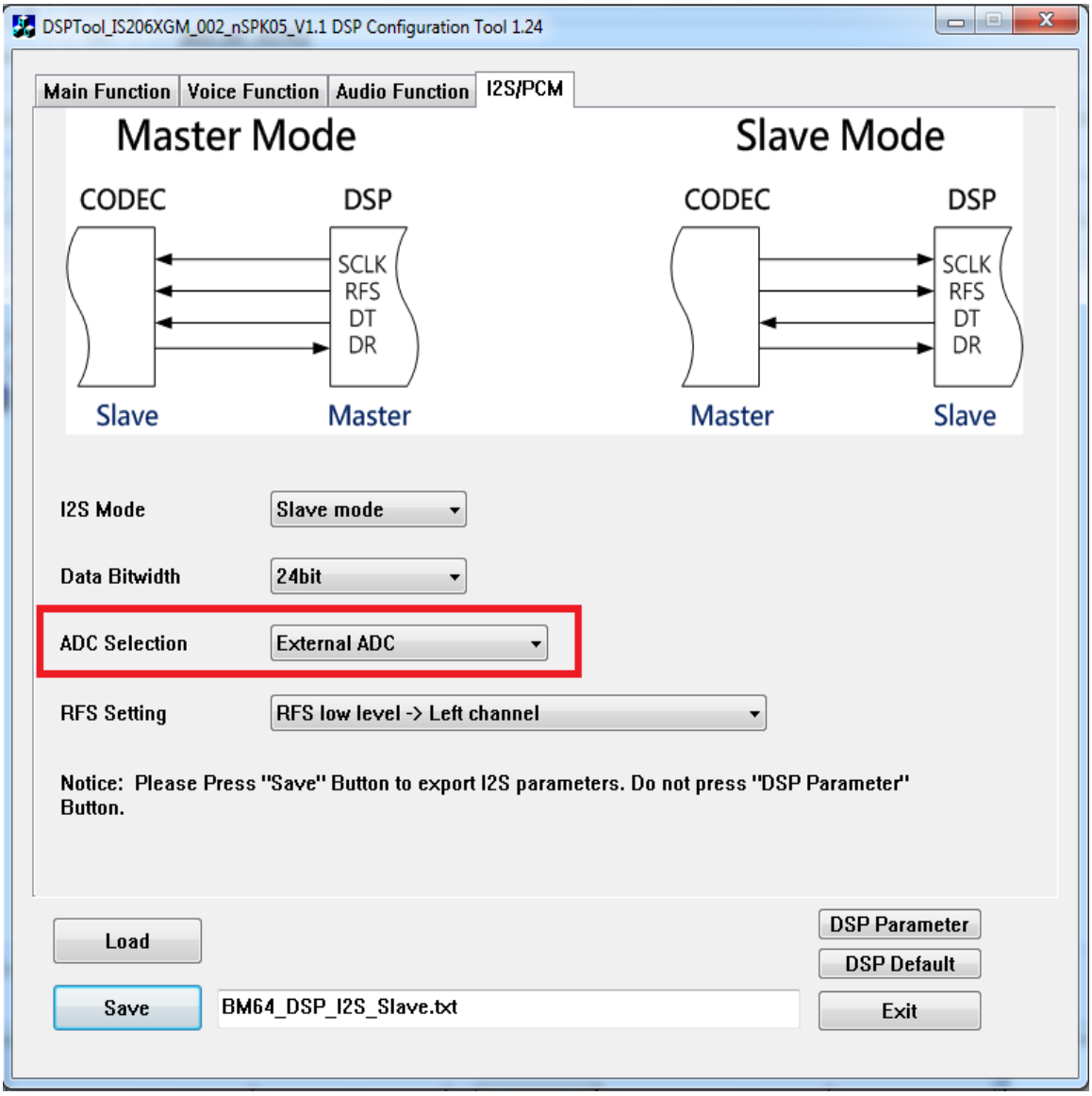


3. **I2S in and I2S out**

Please choose the following in UI tool

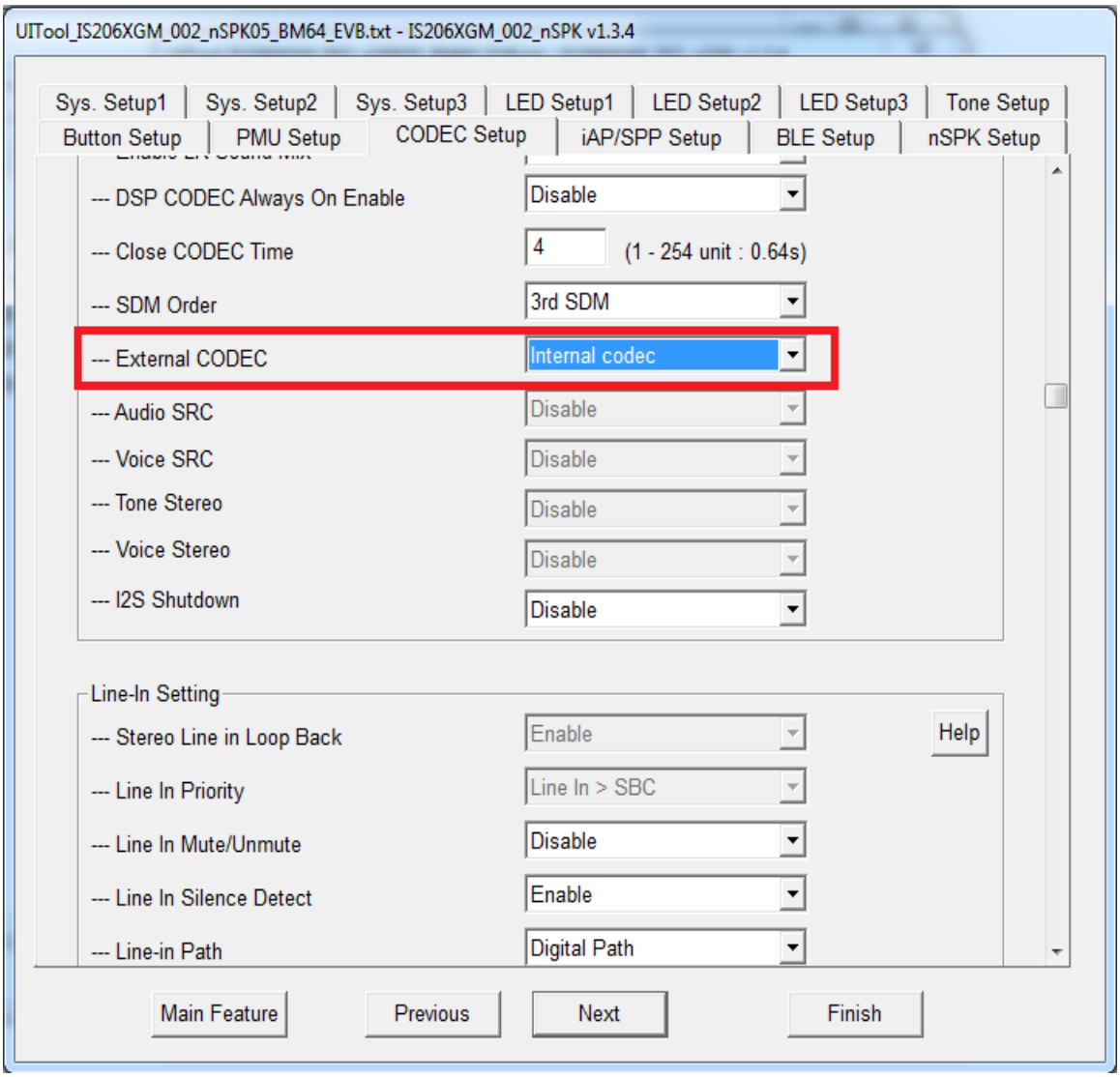


And choose following in DSP Tool

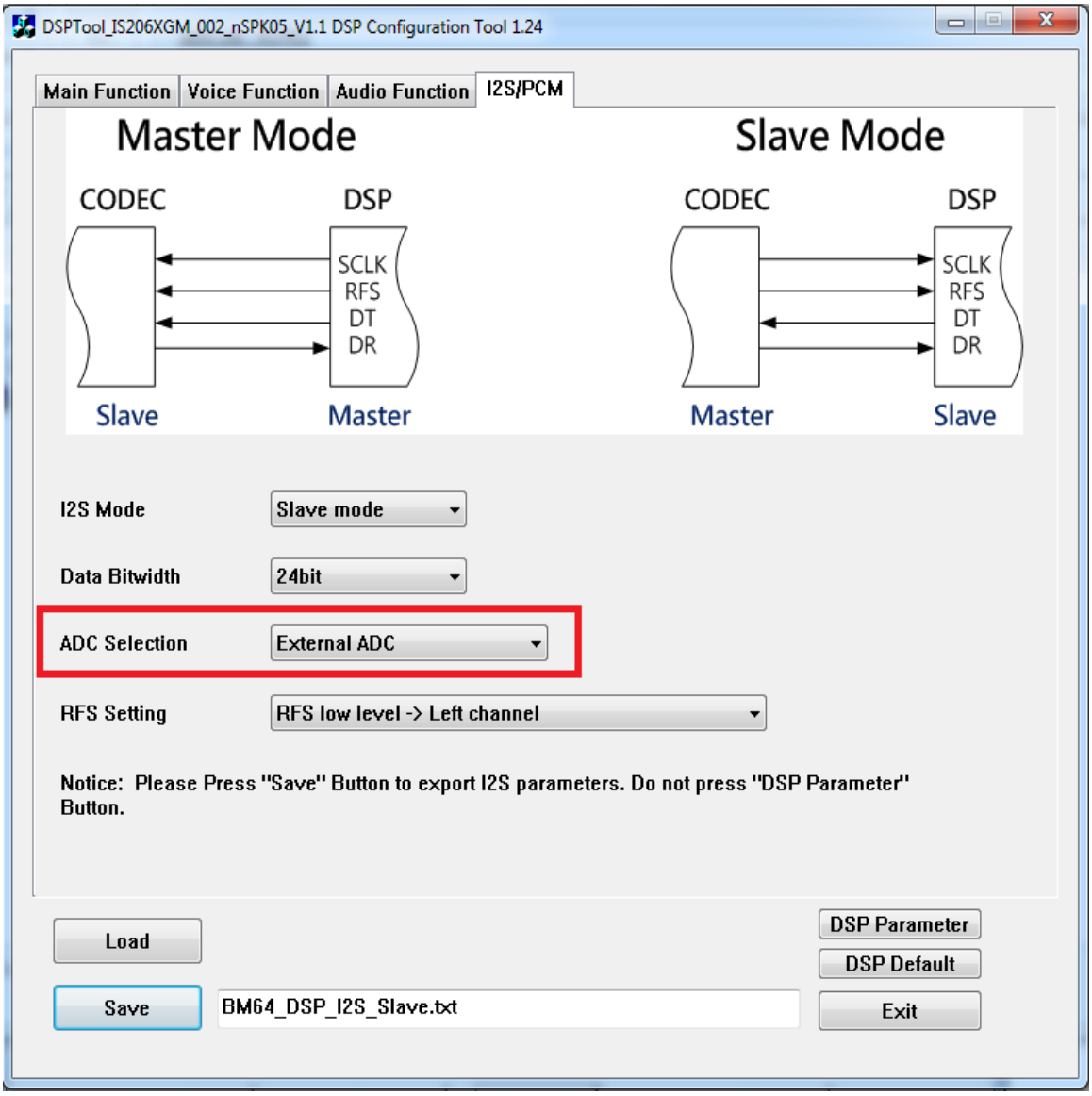


4. I2S in and Analog out

For this configuration choose the following in the UI tool



And chose the following in DSP Tool



Create *.ipf file and program it into BM64. Since this configuration is not supported in UI/DSP tools, hence the following changes needs to be made in EEPROM location

0x1FC = 0x4A

0x344 = 0x0

0xAA0 = 0x3

0xAB4 = 0x3

Note 1: These address is specific to MSPK EEPROM table. DSPK will have different address.

Note2: When I2S_in is selected then it is advisable to use I2S slave mode.

Testing

First two configurations can be tested easily using one BM64 EVB.

1. Please program attached MCHP_Multi_Analog1.ipf for testing Analog-in/Analog-out configuration



MCHP_Multi_Analog1.ipf

2. Please Program EEPROM with MCHP_MULTI_03.ipf for Analog-in/I2S-out configuration.



MCHP_Multi_03.ipf

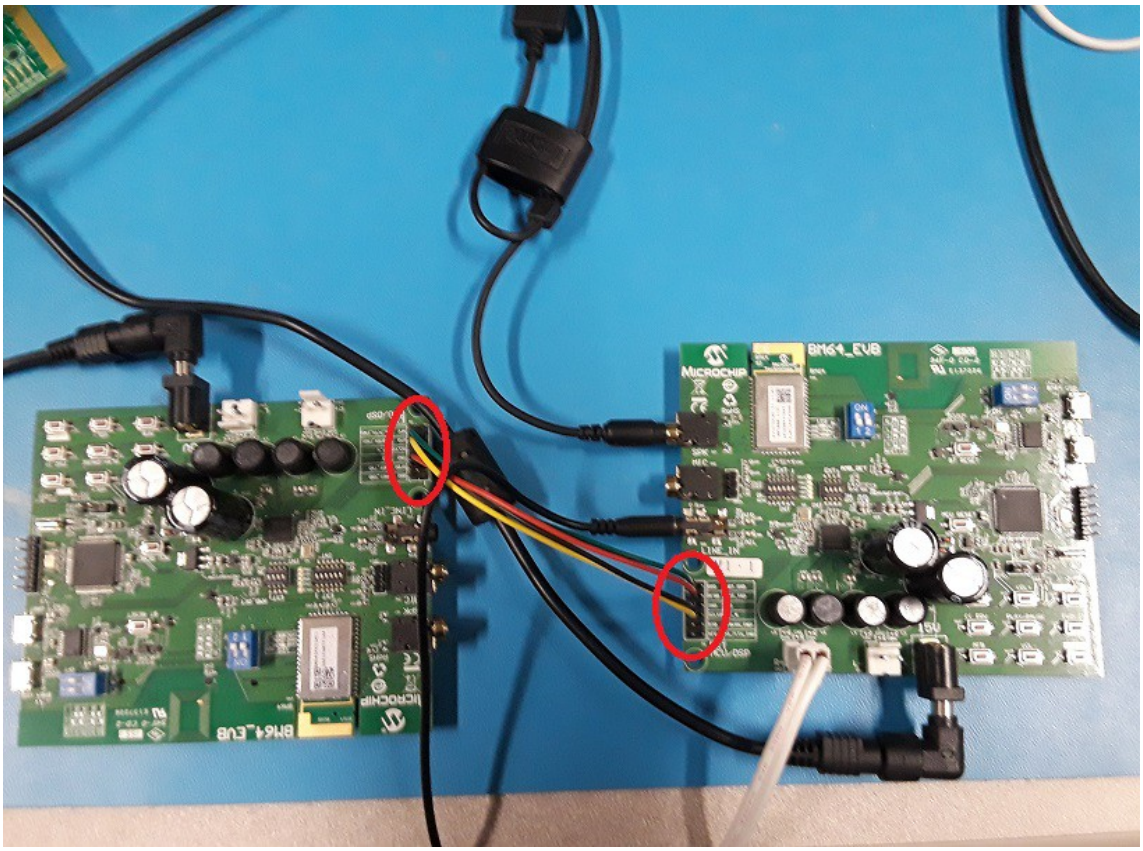
3. For testing I2S-in/I2S-Out configuration, two BM64 EVB is needed. Program Board 1 with MCHP_Multi_03.ipf. Program with Board 2 with MCHP_Multi_I2Sin_I2Sout.ipf



MCHP_Multi_I2Sin_I2sout.ipf

Make the following connections between Jumper J6 of Board 1 and Board 2

BOARD 1	**BOARD 2**
RFSO(3)	RFSO(3)
SCLK0(7)	SCLK0(7)
GND(5)	GND(5)
DT0(9)	DR0(1)



Pair and connect Board with a cell phone and play music. Power on board 2 and connect a speaker with R+/R-. Also insert a cable in line-in. This is to enable line-in detect code in MCU. Music will be heard on connected speaker of Board 2.

4. For testing I2S-in/Analog-Out configuration, two BM64 EVB is needed. Program Board 1 with MCHP_Multi_03.ipf. Program with Board 2 with MCHP_Multi_I2Sin_Analogout.ipf. Make the following changes in EEPROM of the board 2 using MicrochipBTUtil.7Z.

MCHP_Multi_I2Sin_Analogout.ipf

0x1FC = 0x4A
0x344 = 0x0
0xAA0 = 0x3
0xAB4 = 0x3

MicrochipBTUtil.7z

Make the following connections between Jumper J6 of Board 1 and Board 2

BOARD 1 **BOARD 2**

RFSO(3)←----------→RFSO(3)
SCLK0(7)←----------→SCLK0(7)
GND(5)←----------→GND(5)
DT0(9)←----------→DR0(1)

Pair and connect Board with a cell phone and play music. Power on board 2 and connect a speaker with SPK. Also insert a cable in line-in. This is to enable line-in detect code in MCU. Music will be heard on connected speaker of Board 2.